

TUTORIAL | DECEMBER 11 2023

Synchronization of dissipatively coupled oscillators

Chenyang Lu ; Mun Kim ; Ying Yang ; Y. S. Gui  ; C.-M. Hu *J. Appl. Phys.* 134, 221101 (2023)<https://doi.org/10.1063/5.0179159>

Articles You May Be Interested In

Generalized N -rotor problems, synchronized subsystems, and associated solitons*Chaos* (August 2024)

How to grow an oscillators' network with enhanced synchronization

Chaos (March 2023)

Properties of zero-lag long-range synchronization via dynamical relaying

Chaos (March 2010)

AIP Advances

Why Publish With Us?

**21DAYS**
average time
to 1st decision**OVER 4 MILLION**
views in the last year**INCLUSIVE**
scope[Learn More](#)

Synchronization of dissipatively coupled oscillators

Cite as: J. Appl. Phys. **134**, 221101 (2023); doi: [10.1063/5.0179159](https://doi.org/10.1063/5.0179159)

Submitted: 29 September 2023 · Accepted: 13 November 2023 ·

Published Online: 11 December 2023



Chenyang Lu,  Mun Kim,  Ying Yang,  Y. S. Cui, ^{a)}  and C.-M. Hu ^{b)} 

AFFILIATIONS

Department of Physics and Astronomy, University of Manitoba, Winnipeg, Manitoba R3T 2N2, Canada

^{a)}Author to whom correspondence should be addressed: Yongsheng.Cui@umanitoba.ca

^{b)}Electronic mail: hu@physics.umanitoba.ca

ABSTRACT

Synchronization is common in both nature and human physiology, often used to illustrate nonlinear dynamics. Interestingly, one can initiate their comprehension of this phenomenon from pure linear systems. In this Tutorial, we begin with a theoretical exploration of coupled oscillators' dynamic behavior, enabling us to discern and contrast the unique attributes of dissipative coupling as opposed to commonly observed coherent coupling. We then examine synchronization in two dissipative coupled linear systems: one with two pendulums mutually linked via the Lenz effect and the other with two RLC oscillators coupled via a resistor. This Tutorial is designed to serve as a concise starting point for researchers interested in exploring synchronization phenomena using a simplified model driven solely by dissipative coupling.

Published under an exclusive license by AIP Publishing. <https://doi.org/10.1063/5.0179159>

I. INTRODUCTION

Synchronization is ubiquitous in both nature and human physiology.^{1–5} Examples include circadian rhythms,^{6–8} physiological rhythms,^{9–11} Josephson junctions,^{12–14} spin-torque nano-oscillators,^{15–17} nanomechanical oscillators,^{18–20} and power-grid networks.^{21–23} In 2020, researchers observed quantum-level synchronization for the first time.²⁴ Generally speaking, synchronization describes a complex dynamical process in a coupled oscillator system, in which the oscillators are locked to a common frequency despite the differences in the natural frequency of individual oscillators. It has led to broad applications; for example, synchronization in laser arrays plays an important role in reaching the high output power.^{25–27}

The relevant factors that affect synchronization include self-oscillators, non-isochronism, non-identity of the oscillators, and coupling mechanisms.^{4,28} Therefore, synchronization is often used as an example to illustrate the importance of understanding nonlinear phenomena. However, theoretical studies indicate that isochronous oscillators could also be synchronized through dissipative coupling,^{4,29,30} which reduces the subsequent mathematical discussion to simple linear ideas. Here, the coupling between oscillators is caused by dissipative dynamics: As an illustration (shown in the right panel of Fig. 1) in the coupled pendulum context, the coupling force (e.g., friction force) depends on the difference of the velocities, instead of the difference in the displacements in coherent

spring-coupled pendulums. The physics of dissipative dynamics is currently of great interest.^{31,32} For example, dissipative couplings have been characterized as the damping-torque in coupled magnetic bilayers^{33–36} and the cavity Lenz effect in coupled magnon-photon systems.³⁷ Such a dissipative coupling changes the interaction Hamiltonian. It is manifested as energy level attraction, with a coalescence of hybridized modes,^{32,34,37–40} which is distinctly different from level repulsion with mode avoided crossing caused by coherent coupling.

Historically, Winfree⁶ intuitively recognized that synchronization in a dissipative system can be characterized solely by phases of the oscillators. Inspired by Winfree's pioneering work, Kuramoto⁴¹ shows that long-term dynamics in a system with dissipative nature can be described by the following governing equations:

$$\dot{\theta}_i = \omega_i + \frac{K}{N} \sum_{j=1}^N \sin(\theta_j - \theta_i), \quad i = 1, \dots, N, \quad (1)$$

where the system is composed of N oscillators with natural frequencies ω_i , phases θ_i , and the coupling rate K (a real positive number). It is convenient to imagine the phases as a large number of points orbiting the unit circle in a complex plane to

22 June 2026 03:10:54

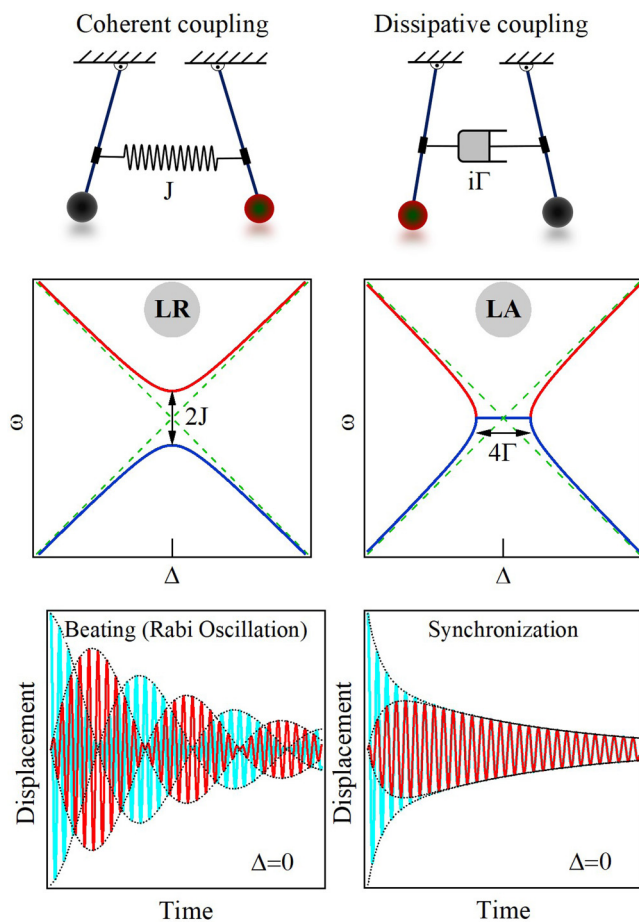


FIG. 1. Left/right panel is for coherent/dissipative coupled pendulums, characterized by level repulsion (LR)/level attraction (LA) dispersion (solid lines) in comparison to the oscillation frequency of uncoupled modes (dashed lines). Time evolution, respectively, shows the beating pattern and synchronization for them. Δ denotes the frequency detuning between two pendulums, and J and Γ , respectively, denote the strength of coherent coupling and dissipative coupling. The damping rates of the two pendulums are set to be the same.

visualize such cooperative dynamics. Here, the average radius r measures the phase coherence, and Ψ is the average phase.^{2,42} If the phases are just randomly distributed, then $r \sim 0$. Following this approach, Eq. (1) becomes $\dot{\theta}_i = \omega_i + K \sin(\Psi - \theta_i)$. The effective interaction, proportional to the system's coherence between the oscillators, pulls θ_i toward the average phase of all involved oscillators. This effect creates a positive feedback loop between coupling and coherence. The effective interaction will increase as the entire system becomes more coherent, attracting more oscillators into synchronization.² This so-called Kuramoto model demonstrates that with a sufficient coupling strength, an enormous system of nearly identical oscillators can be locked to a common frequency by the coupling between them.^{2,42} It is important to recognize that this coupling strength is influenced by a

complex interplay of factors, including network topology, coupling function, natural frequencies, noise, dissipative effects, and various other system-specific parameters. In this Tutorial, our focus remains on the simplest case of synchronization, driven by pure dissipative coupling. For more in-depth insights, the readers can refer to reviews such as those found in Refs. 2 and 43.

Mathematically, dissipative and coherent couplings are equivalent under frequency and damping exchange in the complex frequency plane,³⁹ and both of them can produce synchronization. However, the physical phenomena associated with the two types of interactions are very different. As shown in Fig. 1, coherent coupling tends to break the modes' degeneracy of the original Hamiltonian, and the scale of the degeneracy breaking (characterized by Rabi oscillation) is set by the coupling strength. Therefore, the involved oscillators are often non-isochronous, and synchronization can be achieved through nonlinear frequency pulling.^{30,44} On the contrary, dissipative coupling tends to establish the modes' degeneracy of the original Hamiltonian, which involves the oscillators that are initially detuned, and the coupling strength determines the maximum oscillation frequency range that can synchronize. Accompanied by the hybridized frequency attraction, the line widths are repelled for dissipative coupling. Consequently, in the time-domain measurement, the heavily damped eigenstate (anti-phase-like state) quickly decay, leading to a synchronized state of two oscillators oscillating in-phase-like after a short time.^{4,30,44,45} However, this effect does not apply to coherent coupling. In the case of coherent coupling, the linewidth attraction leads to a similar decay rate of the two hybridized modes. As a consequence, time-domain measurements reveal a beat pattern, as depicted in the lower left panel of Fig. 1.

In the left panel of Fig. 1, we have a textbook example of coherently coupled systems, namely, spring-coupled pendulums. These systems are often used to illustrate the characteristics of coherent coupling due to their intuitive nature. In contrast, dissipative coupling is less commonly encountered in various physical scenarios. To bridge the gap and make the topic accessible to a broad audience, including non-experts, this Tutorial begins by briefly introducing the motion equations of coupled harmonic oscillators. We then employ the method of averaging to derive phase evolution for a dissipatively coupled system and link it with the general relation given by Eq. (1), based on the Kuramoto model. Drawing insights from these studies, we examine the dynamics of two dissipatively coupled systems: one involving two pendulums mutually linked via the Lenz effect and the other featuring two RLC oscillators coupled by a resistor. Through the measured time evolution of oscillator amplitudes, we illustrate the distinct features inside and outside the synchronization zone. These systems can serve as an educational tool for both graduate and undergraduate students. For a deeper comprehension of synchronization, we suggest referring to the following textbooks and reviews: Refs. 2, 4, 5, 28, 29, and 43.

II. MATHEMATICAL FORMULATION OF SYNCHRONIZATION DRIVEN BY DISSIPATIVE COUPLING

In this section, we investigate a canonical system with two coupled harmonic oscillators, denoted as variables x_1 and x_2 . To

22 June 2026 03:10:54

simplify the exposition and avoid delving into complex nonlinear mathematics, this Tutorial focuses exclusively on the characteristics of a linear coupled system. Even in the absence of any nonlinear ingredients, such as the interplay between nonlinear damping and gain resulting in self-sustained oscillation, these two oscillators can achieve synchronization through dissipative coupling. Our initial analysis examines the coupled system's dynamics, as governed by the following equations of motion:

$$\ddot{x}_1 + 2\gamma_1 \dot{x}_1 + \omega_1^2 x_1 = f_1(x_1, \dot{x}_1) + 2J_1 \omega_1 (x_2 - x_1) + 2\Gamma_1 (\dot{x}_2 - \dot{x}_1), \quad (2a)$$

$$\ddot{x}_2 + 2\gamma_2 \dot{x}_2 + \omega_2^2 x_2 = f_2(x_2, \dot{x}_2) + 2J_2 \omega_2 (x_1 - x_2) + 2\Gamma_2 (\dot{x}_1 - \dot{x}_2). \quad (2b)$$

The LHS of equations includes the linear restoring force with frequency $\omega_{1,2}$ and the linear frictional force with damping coefficient $\gamma_{1,2}$. The $f_{1,2}$ terms include nonlinear restoring force and nonlinear friction determined by non-isochronous and self-oscillation effects. Coupling terms in RHS are chosen to be proportional to the difference of the variables and their derivatives and vanishes if the states of the two subsystems coincide, i.e., $x_1 = x_2$ and $\dot{x}_1 = \dot{x}_2$. The coupling constants $J_{1,2}$ and $\Gamma_{1,2}$, respectively, describe the coherent and dissipative interaction between the oscillators. For brevity and simplicity, setting $J_1 = J_2$ and $\Gamma_1 = \Gamma_2$ is a common practice. However, it is important to note that in reality, these coupling constants can vary, as demonstrated by the derived equation of motion for dissipatively coupled pendulums in [Appendix B](#).

Complete analytical solutions are not always available for this coupled system described by Eq. (2). However, in a situation $\gamma_{1,2} \ll J_{1,2}$, $\Gamma_{1,2} \ll \omega_{1,2}$, analytical expressions may be derived by appropriate approximations. We first simplify Eq. (2) in a rotating reference frame with a reference frequency $\omega_{ref} = (\omega_1 + \omega_2)/2$ and a slowly varying envelope function $a_{1,2}(t)$ such that $x_{1,2} = [a_{1,2}(t)e^{i\omega_{ref}t} + a_{1,2}^*(t)e^{-i\omega_{ref}t}]/2$. For a linear dissipative coupled system, where $f_1 = f_2 = 0$ and $J_1 = J_2 = 0$, we obtain

$$\frac{d}{dt} \begin{pmatrix} a_1 \\ a_2 \end{pmatrix} = i \begin{pmatrix} -\Delta/2 + i(\gamma_1 + \Gamma_1) & -i\Gamma_1 \\ -i\Gamma_2 & \Delta/2 + i(\gamma_2 + \Gamma_2) \end{pmatrix} \begin{pmatrix} a_1 \\ a_2 \end{pmatrix}. \quad (3)$$

Here, $\Delta = \omega_2 - \omega_1$ denotes frequency detuning.

In the rotating reference frame, the complex eigenfrequencies of the hybridized modes, $\tilde{\omega}_{\pm}$, deduced from Eq. (3) are

$$\tilde{\omega}_{\pm} - \omega_{ref} = \frac{i(\gamma_1 + \Gamma_1 + \gamma_2 + \Gamma_2)}{2} \pm \frac{1}{2} \sqrt{[\Delta + i(-\gamma_1 - \Gamma_1 + \gamma_2 + \Gamma_2)]^2 - 4\Gamma_1\Gamma_2}. \quad (4)$$

The real and imaginary parts of $\tilde{\omega}_{\pm}$, respectively, represent the eigenfrequency and linewidth of the hybridized mode.

The dispersion and linewidth evolution calculated from Eq. (4) are plotted in [Fig. 2](#). Distinctly different from level repulsion

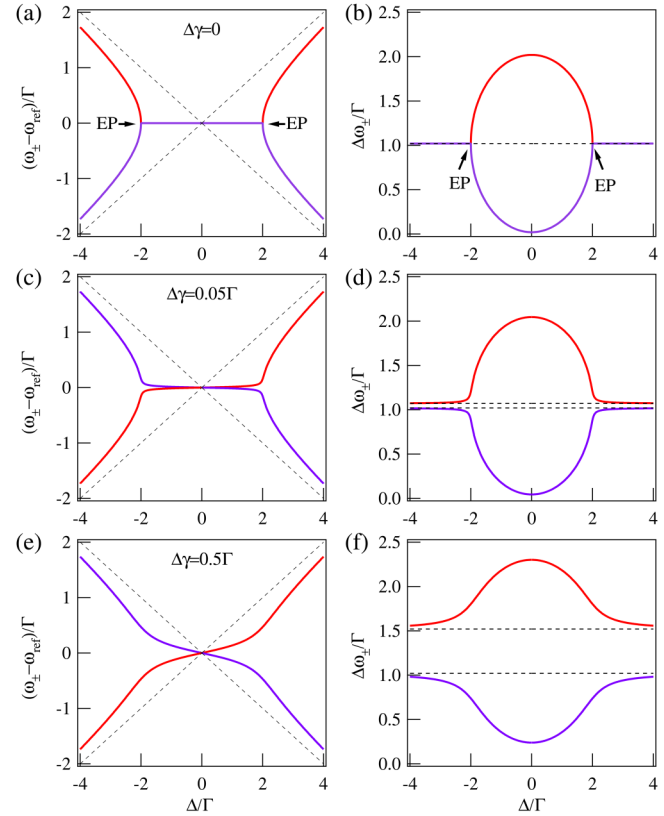


FIG. 2. (a), (c), and (e) Hybridized frequency $\omega_{\pm}$ [rotating with a frequency $\omega_{ref} = (\omega_1 + \omega_2)/2$] and (b), (d), and (f) linewidth $\Delta\omega_{\pm}$ of a dissipatively coupled system calculated according to Eq. (4). Red (blue) curves correspond to the heavily damped ω_{+} (lightly damped ω_{-}) branch. The diagonal dashed lines in (a), (c), and (e) show the frequencies of the uncoupled pendulums, and horizontal dashed lines in (b), (d), and (f) correspond to $\Gamma + \gamma_1$ and $\Gamma + \gamma_2$. Arrows in (a) and (b) indicate the exceptional points (EPs). Throughout these figures, $\omega_1/2\pi = 1$ Hz, $\Gamma_1 = \Gamma_2 = \Gamma = 0.025\omega_1$, and $\gamma_1 = 0.02\Gamma$. $\Delta\gamma = \gamma_2 - \gamma_1 = 0$ for (a) and (b), $\Delta\gamma = 0.05\Gamma$ for (c) and (d), and $\Delta\gamma = 0.5\Gamma$ for (e) and (f).

with mode anticrossing caused by coherent coupling, level attraction is observed. In the meantime, the line widths, $\Delta\omega_{\pm} = \Im m(\tilde{\omega}_{\pm})$, of the hybridized modes, determined by the imaginary eigenspectrum, are repelled.

In physics, the fundamental distinction between coherent coupling and dissipative coupling lies in their energy conservation properties. Dissipative coupling specifically refers to the loss or dissipation of energy in a system due to the interaction or coupling between its components. Mathematically, this difference is often expressed through the coupling's dependence on displacement ($x_{1,2}$) or velocity ($\dot{x}_{1,2}$). While coherent coupling facilitates the coherent exchange of energy between the coupled subsystems, dissipative coupling tends to attain a stable equilibrium state by dissipating energy through a heavily damped process.

A. Dynamic states of an anti- $\mathcal{PT}$ symmetric system

More interestingly, both eigenfrequencies and linewidths of the two hybridized modes can exactly coalesce at a critical condition indicated by arrows in Figs. 2(a) and 2(b). These spectral singularities in the parameter space of a coupled system are named as exceptional points (EPs).^{46–49}

With these observations as shown in Figs. 2(a) and 2(b), an anti- $\mathcal{PT}$ symmetric system can be constructed at $\Gamma_1 = \Gamma_2 = \Gamma$ and $\gamma_1 = \gamma_2 = \gamma$. Now the system has a characteristic Hamiltonian expressed as $\hat{H} = \begin{pmatrix} -\Delta/2 + i(\gamma + \Gamma) & -i\Gamma \\ -i\Gamma & \Delta/2 + i(\gamma + \Gamma) \end{pmatrix}$, which satisfies anti- $\mathcal{PT}$ symmetry,

$$\{\hat{H}, \hat{P}\hat{T}\} = 0. \quad (5)$$

Here, $\hat{P} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ represents the parity operator, and $\hat{T}$ represents the time-reversal operator, which transforms the temporal parameter $t \mapsto -t$, i.e., $i \mapsto -i$ in the Hamiltonian. In this case, the complex eigenfrequencies of the hybridized modes are

$$\tilde{\omega}_{\pm} - \omega_{ref} = i(\gamma + \Gamma) \pm \sqrt{(\Delta/2)^2 - \Gamma^2}, \quad (6)$$

and the corresponding eigenvectors are

$$\begin{pmatrix} 1 \\ i \left[\Delta/2\Gamma \pm \sqrt{(\Delta/2\Gamma)^2 - 1} \right] \end{pmatrix}. \quad (7)$$

At zero detuning ($\Delta = 0$), the eigenvector of the ω_+ mode is $\begin{pmatrix} 1 \\ -1 \end{pmatrix}$, which corresponds to the anti-phase state, and the eigenvector of the ω_- mode is $\begin{pmatrix} 1 \\ 1 \end{pmatrix}$, which corresponds to the in-phase state.

As shown in Figs. 2(a) and 2(b), the two hybridized modes have different oscillation frequencies but with the same damping rate for the broken anti- $\mathcal{PT}$ -symmetry phase ($|\Delta| > 2\Gamma$). In contrast, under the condition of the exact anti- $\mathcal{PT}$ -symmetry phase ($|\Delta| < 2\Gamma$), the two hybridized modes have the same oscillation frequency but with different damping rates. At EPs ($\Delta = \pm 2\Gamma$),

the complex eigenfrequencies in Eq. (4) exactly degenerate, i.e., $\tilde{\omega}_+^{EP+} \equiv \tilde{\omega}_-^{EP+} = \omega_1 + \Gamma + i(\gamma + \Gamma)$ and $\tilde{\omega}_+^{EP-} \equiv \tilde{\omega}_-^{EP-} = \omega_1 - \Gamma + i(\gamma + \Gamma)$. The eigenvectors $\begin{pmatrix} 1 \\ \pm i \end{pmatrix}$ will also merge and degenerate at EPs, which serve as branch point singularities that separate the equatorial state and the meridian state in the Bloch sphere. The distinct degeneracy feature of a dissipative coupled system with an anti- $\mathcal{PT}$ symmetry is summarized in Table I.

B. Time evolution of an anti- $\mathcal{PT}$ symmetric system

Having realized the eigenfrequencies and eigenvectors of the system can coincide exactly with each other, we now focus on phase locking of the two oscillators due to dissipative coupling. To investigate the stationary states, our dynamical system can be re-expressed as $x_{1,2} = A_{1,2}(t)\cos[\omega_{ref}t + \theta_{1,2}(t)]$, which contains time scales separation: a fast oscillation $\cos(\omega_{ref}t)$ vs a slow drift in the amplitude $A_{1,2}(\geq 0)$ and phase $\theta_{1,2}$. As usually done for such dynamical systems, the method of averaging^{4,28} is used, in which we average the variables $x_{1,2}$ over the oscillation period to eliminate fast oscillations and observe the qualitative behavior of $A_{1,2}$ and $\theta_{1,2}$. Through this mathematical process (discussed in more detail in Appendix A), we get the phase correlation between the two oscillators,

$$\frac{d(\theta_2 - \theta_1)}{dt} = \Delta - \left(\frac{A_1}{A_2} + \frac{A_2}{A_1} \right) \Gamma \sin(\theta_2 - \theta_1). \quad (8)$$

This equation looks similar to the simplest case of the Kuramoto model described by Eq. (1). They all show the competition mechanism between frequency detuning and coupling term (pure sine wave) in order to achieve synchronization. Without coupling, the phase difference of the two oscillators increases linearly at a constant rate of Δ . However, if the coupling strength is sufficiently large; the frequency detuning can be completely compensated, producing synchronization of the two oscillators, i.e., $d(\theta_2 - \theta_1)/dt = 0$.

It is important to note that while the derivation from Eqs. (2) to (8) offers an educational tutorial on the synchronization of two dissipatively coupled oscillators, it simplifies and omits the entirety of the stochastic nature inherent in the Kuramoto equation and its associated rich physics. Thus, to develop a comprehensive

TABLE I. Key features of a dissipative coupled system with anti- $\mathcal{PT}$ symmetry. Here, $\Theta = \arcsin(\Delta/2\Gamma)$ for $|\Delta| < 2\Gamma$ and $r_{\pm} = \Delta/2\Gamma \pm \sqrt{(\Delta/2\Gamma)^2 - 1}$ for $|\Delta| > 2\Gamma$.

	Between two EPs	At two EPs	Outside two EPs
Frequency range	$ \Delta < 2\Gamma$	$\Delta = \pm 2\Gamma$	$ \Delta > 2\Gamma$
Eigenfrequency	$\Re(\tilde{\omega}_+) = \Re(\tilde{\omega}_-)$	$\Re(\tilde{\omega}_+) = \Re(\tilde{\omega}_-)$	$\Re(\tilde{\omega}_+) \neq \Re(\tilde{\omega}_-)$
Linewidth	$\Im(\tilde{\omega}_+) \neq \Im(\tilde{\omega}_-)$	$\Im(\tilde{\omega}_+) = \Im(\tilde{\omega}_-)$	$\Im(\tilde{\omega}_+) = \Im(\tilde{\omega}_-)$
Eigenvector	$[1, \mp \exp(\mp i\Theta)]$	$[1, \text{sgn}(\Delta)i]$	$[1, r_{\pm}i]$
Time domain characteristic	Two tone decay producing synchronization	Critical transition point	Two tone oscillations producing beat-like pattern
Synchronization	With a phase difference Θ between two oscillators	Critical transition point	Never

understanding of synchronization across diverse coupled systems, one must account for the stochastic effects.

Using constant phase difference $\Theta = \theta_2 - \theta_1$, we can solve the motion equation [Eq. (A3) in Appendix A] with the initial conditions $x_1(t=0) = x_0$, $x_2(t=0) = 0$ and get

$$x_1 = x_0 e^{-(\gamma+\Gamma)t} (e^{\Gamma t \cos \Theta} + e^{-\Gamma t \cos \Theta}) \cos(\omega_{\text{ref}} t)/2, \quad (9a)$$

$$x_2 = x_0 e^{-(\gamma+\Gamma)t} (e^{\Gamma t \cos \Theta} - e^{-\Gamma t \cos \Theta}) \cos(\omega_{\text{ref}} t + \Theta)/2. \quad (9b)$$

The effect of dissipative coupling is twofold: on the one hand, in addition to intrinsic damping, it acts as an additional dissipation channel, expressed as $e^{-\Gamma t}$; on the other hand, it can introduce a motion component with effective gain ($\Gamma \cos \Theta$) and another motion component with effective loss ($-\Gamma \cos \Theta$), as clearly shown in brackets. Obviously, the contribution of the second component, after a short period of time, becomes negligible. As a result, the two oscillators will decay with the same rate and oscillate at the same frequency (but with a phase difference Θ between them).

Note that in Eqs. (9a) and (9b), the sign for the first component is the same, while the sign for the second component is the opposite, indicating their in-phase/anti-phase motion nature. For the simplest case at $\Theta = 0$, the first component leads to the in-phase motion between x_1 and x_2 (corresponding to zero friction), and the second component moves the two oscillators in opposite directions (corresponding to maximum friction). Therefore, dissipative coupling caused by the nature of friction tends to produce in-phase synchronization. If $\Gamma \cos \Theta \gg \gamma$, the dynamics of two dissipative coupled oscillators follows a two-tone decay process: a quick decay of anti-phase-like motion producing synchronization and a slow decay of in-phase-like motion.

Substituting constant $A_1/A_2 = 1$ (a more general solution will be provided in Sec. II C) into Eq. (8), we get a simple relation among the phase difference, the coupling strength, and detuning expressed as

$$\Theta = \arcsin(\Delta/2\Gamma). \quad (10)$$

With this relation, the eigenvectors [Eq. (7)] discussed in Sec. II A can be simplified as $\begin{pmatrix} 1 \\ \mp e^{\mp i\Theta} \end{pmatrix}$, corresponding to $\tilde{\omega}_{\pm}[\omega_{\text{ref}} + i(\gamma + \Gamma) \pm i\Gamma \cos \Theta]$ eigenstates. Then, in the complex plane, the dynamic equations (9a) and (9b) can be re-expressed as the superposition of two eigenstates, expressed as

$$\begin{pmatrix} \tilde{x}_1 \\ \tilde{x}_2 \end{pmatrix} \propto \begin{pmatrix} 1 \\ e^{i\Theta} \end{pmatrix} e^{\tilde{\omega}_- t} + \begin{pmatrix} 1 \\ -e^{-i\Theta} \end{pmatrix} e^{\tilde{\omega}_+ t}. \quad (11)$$

Since the ω_+ state is heavily damped [see Fig. 2(b)], the dynamics of the two coupled oscillators after a short period is dominated by the ω_- state. In this way, the phase difference Θ between x_1 and x_2 in the time-domain measurement can be used to directly monitor the hybridized states in the Bloch sphere.

Table I contains a summary of the key characteristics for an anti- $\mathcal{PT}$ symmetric system (1) between two EPs (the anti- $\mathcal{PT}$ unbroken phase), (2) at two EPs, and (3) outside two EPs (the anti- $\mathcal{PT}$ broken phase).

C. Dynamic states and time evolution for $\gamma_1 \neq \gamma_2$

In this subsection, we will briefly discuss the dynamic state and time evolution of two dissipatively coupled oscillators with uneven damping rates (i.e., $\Delta\gamma = \gamma_2 - \gamma_1 \neq 0$). It remains that the motion of the two oscillators at $\Delta = 0$ can be either in-phase or anti-phase because according to Eq. (8), $\sin \Theta = 0$. Except for this point, the square root term in Eq. (4) is no longer a pure imaginary number, so the frequencies of the two mixed modes can only coincide at $\Delta = 0$ [Figs. 2(c) and 2(e)].

For simplicity, we set $\Gamma_1 = \Gamma_2 = \Gamma$. Damping repulsion produced by dissipative coupling creates a forbidden linewidth gap between $\gamma_1 + \Gamma$ and $\gamma_2 + \Gamma$, which is more clearly seen in Figs. 2(d) and 2(f). At $\Delta = 0$ and $\Gamma \gg |\Delta\gamma|$, the minimum damping state [with a damping rate of $\sim (\gamma_1 + \gamma_2)/2$] corresponds to the in-phase motion of the coupled system, and therefore, the dissipation caused by the coupling effect is zero. On the contrary, the maximum damping state [with a damping rate of $\sim 2\Gamma + (\gamma_1 + \gamma_2)/2$] corresponds to the anti-phase motion.

Following the similar mathematical process discussed in Sec. II B, we can get

$$x_1 = x_0 e^{-[\Gamma + (\gamma_1 + \gamma_2)/2]t} (C_{11} e^{\Gamma' t} + C_{12} e^{-\Gamma' t}) \frac{\cos(\omega'_{\text{ref}} t)}{4\Gamma'}, \quad (12a)$$

$$x_2 = x_0 e^{-[\Gamma + (\gamma_1 + \gamma_2)/2]t} (C_{21} e^{\Gamma' t} - C_{22} e^{-\Gamma' t}) \frac{\cos(\omega'_{\text{ref}} t + \Theta)}{4\Gamma'}. \quad (12b)$$

Here, $\Gamma' = \sqrt{\Gamma^2 \cos^2 \Theta + \Delta\gamma^2/4}$ is the effective dissipation rate by considering the uneven damping rate, the phase shift between two oscillators, and the dissipative coupling strength, which corresponds to the half interval between the linewidths of the hybridized mode. Other coefficients are $C_{11} = 2\Gamma' + \Delta\gamma$, $C_{12} = 2\Gamma' - \Delta\gamma$, and $C_{21} = C_{22} = 2\Gamma \cos \Theta$. Except for specific coefficients, Eqs. (12) and (9) have the same form, indicating that they have the same physical origin.

It should be noted that in the steady-state, it means when $d\Theta/dt = 0$, but it is not necessary when $d\theta_1/dt = d\theta_2/dt = 0$. Because of $\Delta\gamma \neq 0$, the long-term ratio $A_1/A_2 \sim C_{11}/C_{21} = \frac{2\sqrt{\Gamma^2 \cos^2 \Theta + \Delta\gamma^2/4} + \Delta\gamma}{2\Gamma \cos \Theta}$ is no longer equal to the unit and the amplitude of the oscillator with a lower damping rate is greater. This effect results in the frequency of the synchronous oscillator slightly offset from the reference frequency, i.e., $\omega'_{\text{ref}} - \omega_{\text{ref}} = d\theta_1/dt = \Delta \frac{A_2^2 - A_1^2}{2(A_1^2 + A_2^2)} + \mathcal{O}(\Delta^2)$. Under the condition of $\Delta\gamma > 0$, ω'_{ref} is blueshifted with respect to ω_{ref} for $\Delta < 0$ and redshifted for $\Delta > 0$. Compared with Figs. 2(c)–2(f), one can find that ω'_{ref} corresponds to an eigenstate with a lower damping rate. Similarly, the motion caused by the terms C_{12} and C_{22} corresponds to the heavily damped eigenstate, and its frequency shift is opposite to the previous case.

Unlike the anti- $\mathcal{PT}$ system, there is no such critical transition point between the synchronous area and the non-synchronous area. Still, we can calculate constant Θ according to Eq. (8), which is plotted as a function of Δ in Fig. 3 for several $\Delta\gamma$. Obviously, for unevenly damped oscillators, the critical transition becomes softer.

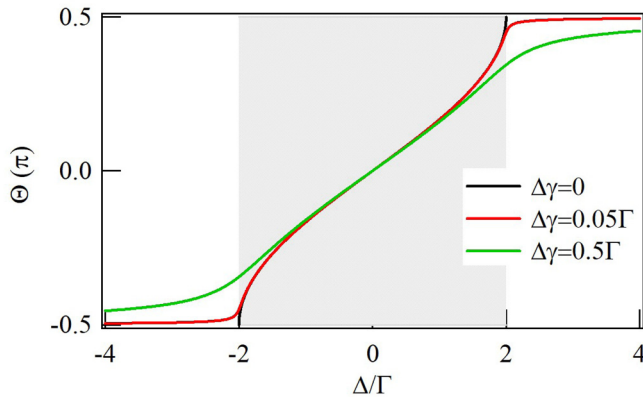


FIG. 3. The phase difference Θ as a function of frequency detuning for $\Delta\gamma = 0$ (black), $\Delta\gamma = 0.05\Gamma$ (red), and $\Delta\gamma = 0.5\Gamma$ (green). $\Delta\gamma = \gamma_2 - \gamma_1$ denotes the damping mismatch between the two oscillators. The gray area indicates the synchronization zone for $\Delta\gamma = 0$, where $\Delta = \pm 2\Gamma$ defines its boundary.

For small detuning, where $\Gamma \cos \Theta \gg |\Delta\gamma|$, the phase Θ is not sensitive to $\Delta\gamma$.

III. SYNCHRONIZATION BASED ON DISSIPATIVELY COUPLED PENDULUMS

Dissipatively coupled pendulums are illustrated in Fig. 4. The pendulums (with equal mass m) are connected to fixed pivot points by two light rods of identical length (about 95 cm). The center of mass of the pendulum 1 is located at a fixed distance of l_1 , while the center of mass of the pendulum 2, denoted as l_2 , can be tuned by moving the counterweight. The coupling accessories [magnets (A) and coil (B) in Fig. 4] are fastened on the pendulums with a distance of l below the pivot points. This setup adopts the Lenz effect to generate mutual dissipative coupling between two pendulums. A more refined approach to construct a laboratory demonstration apparatus can be explored in the Refs. 50 and 51.

According to Lenz's law, the direction of the current induced in the coil due to the movement of the magnet should be able to resist the change of the magnetic flux and exert a mechanical force opposite to the movement of the magnet. Therefore, the coil pendulum is driven to follow the movement of the magnet pendulum to keep the magnetic flux constant. We will show that the two pendulums can be synchronized and move in phase through the mutual dissipative interaction.

We start by analyzing the dissipative coupling in this setup. Lenz's law is usually incorporated into Faraday's law with a negative sign, which stipulates that the induced EMF $\mathcal{E}$ and the rate of change of magnetic flux Φ_B have opposite signs,

$$\mathcal{E} = -d\Phi_B/dt \propto -d[l\sin(\varphi_1) - l\sin(\varphi_2)]/dt \propto \dot{\varphi}_2 - \dot{\varphi}_1. \quad (13)$$

As shown in Fig. 4, the change in the magnetic flux is roughly proportional to the relative position between magnets (A) and coil (B), i.e., $\approx l(\varphi_1 - \varphi_2)$. Therefore, the back EMF is proportional to

the difference of the derivative of the variable $\varphi_{1,2}$, satisfying the definition of dissipative coupling in Eq. (2).

Before the coupling experiment, each pendulum was measured independently to obtain the relevant parameters $\omega_{1,2}$ and $\gamma_{1,2}$. As expected, the oscillator frequencies of the pendulums follow the relation of $\omega_{1,2} = \sqrt{g/l_{1,2}}$ (g , the acceleration of gravity), and we have $\omega_1/2\pi = 0.65$ Hz and $\omega_2/2\pi = (0.57 - 0.87)$ Hz accompanied by $\gamma_1/2\pi = 1.0$ mHz and $\gamma_2/2\pi = (0.7 - 3.1)$ mHz.

We first perform the coupling experiment at $\Delta = 0$ and measure the time evolution of φ_2 [circles in Fig. 5(a)] with an initial condition of $\varphi_1 = \varphi_0$ and $\varphi_2 = 0$. Then, FFT is used to convert the measurement results in the time-domain into a representation in the frequency-domain, as shown in Fig. 5(b). The deduced oscillator frequency of $\omega_{\pm}/2\pi \sim 0.65$ Hz is in agreement with the experimental condition $\omega_1/2\pi = \omega_2/2\pi = 0.65$ Hz. Since $\Delta = 0$, we can directly calculate the time-evolution of φ_2 [solid line in Fig. 5(a)] according to Eq. (12b) with pre-obtained parameters $\gamma_1/2\pi = \gamma_2/2\pi = 1.0$ mHz and adjusting parameter $\Gamma_1/2\pi = \Gamma_2/2\pi = 0.027$ Hz. A similar result was also obtained by numerical simulation based on Eq. (B4). Due to the in-phase motion of the two pendulums, as shown in Fig. 6(a), the net dissipation caused by the coupling effect is zero. The slow decay after a short period of time is solely determined by the intrinsic dampings $\gamma_{1,2}$.

For $\Delta/2\pi = -0.030$ Hz corresponding to frequency mismatch less than $2\Gamma_1$, the time-evolution of φ_2 still looks like a simple damped oscillation and only one FFT peak appears at $\omega_{\pm}/2\pi = 0.634$ Hz [$\approx (\omega_1 + \omega_2)/4\pi$]. These experimental observations are in agreement with the previous discussion in Sec. II.

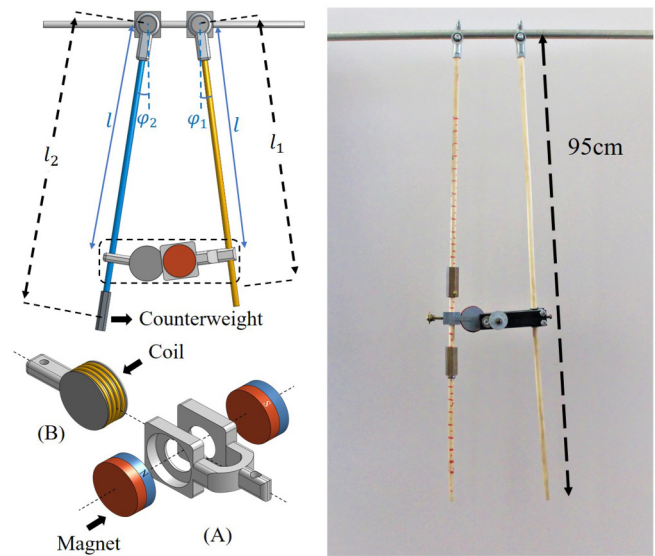


FIG. 4. Pendulums mutually coupled via the Lenz effect. This system is characterized by dissipative coupling of the oscillations, which results in level attraction of the hybridized modes. A movable counterweight is used to match the masses of two pendulums and to tune the oscillator frequency ω_2 . The right panel shows the picture of the actual experimental setup.

22 June 2026 03:10:54

Although both pendulums oscillate at the same frequency, it can be clearly observed that for larger frequency detuning, the oscillation amplitude decays faster (corresponding to a broader FFT peak). This phenomenon is caused by the phase misalignment between the two pendulums [see Fig. 6(a)], as predicated according to Eq. (A3c).

Further increasing the frequency mismatch greater than $2\Gamma_1$ to $\Delta/2\pi = -0.077$ Hz, the time-evolution of φ_2 shows completely different characteristics. The observed beating pattern is caused by two close oscillation frequencies. Through FFT we obtain, their frequencies are $\omega_+/2\pi = 0.644$ Hz and $\omega_-/2\pi = 0.582$ Hz. This phenomenon can be explained by the fact that the coupling strength is not strong enough to compensate for extra dissipation caused by the huge difference in natural oscillation frequency, so the system cannot achieve frequency locking. Compared with the first two cases, the oscillation amplitude decays even faster in this situation. Figure 5(c) summarizes Δ -dependent FFT spectra, where the dashed lines guide the evolution of the FFT peak. The mergence of oscillation peaks is observed in this experiment, accompanied by the narrowing of the FFT peak.

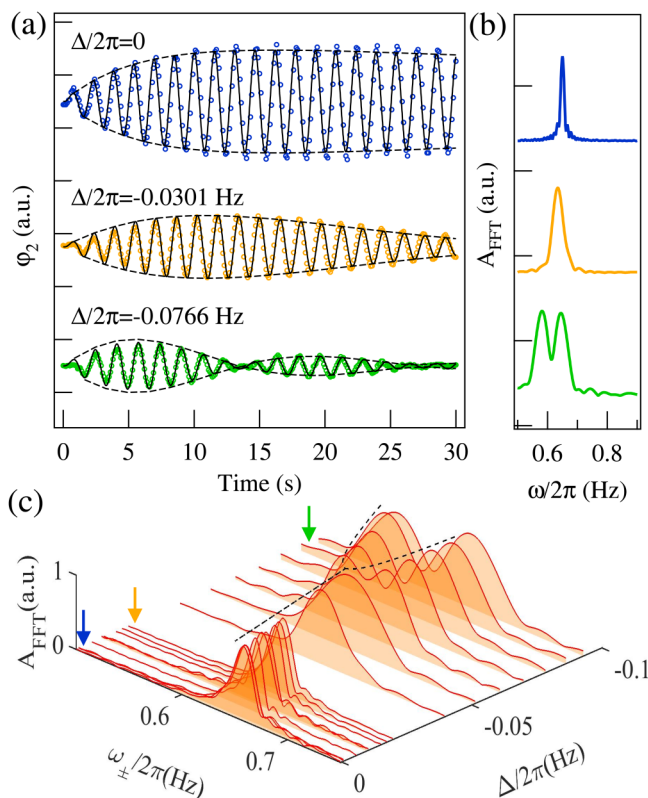


FIG. 5. (a) The time evolution of φ_2 at different detuning, where circles are experimental data, solid lines are simulation results according to Eq. (B4), and dashed lines indicate the envelope of the coupled oscillation. (b) plots the corresponded FFT spectra (normalized). (c) Δ -dependent normalized FFT spectra, where the dashed lines guide the evolution of oscillation frequencies.

In order to understand the characteristic phase correlation between two pendulums during the dynamical process toward synchronization, we have simultaneously measured the time evolution of both pendulums. Figure 6(a) plots $\varphi_{1,2}(t)$ at three typical cases: $\Delta > 0$, $\Delta = 0$, and $\Delta < 0$. In all three cases, the two pendulums oscillate with a common frequency after a short period of time. Then, we carefully inspect the phase correlation between the two pendulums. As expected, two pendulums oscillate in-phase when $\Delta = 0$, and a slight phase shift appears when $\Delta \neq 0$. When $\Delta > 0$, the oscillation of the coil pendulum (blue wave) leads the magnet pendulum (gray wave), and when $\Delta < 0$, it (green wave) lags the magnet pendulum.

To precisely determine the phase difference between them, we use $\varphi_{1,2} \propto e^{-\Lambda t} \cos(\omega t + \theta_{1,2})$ [solid lines in the insets of Fig. 6(a)]

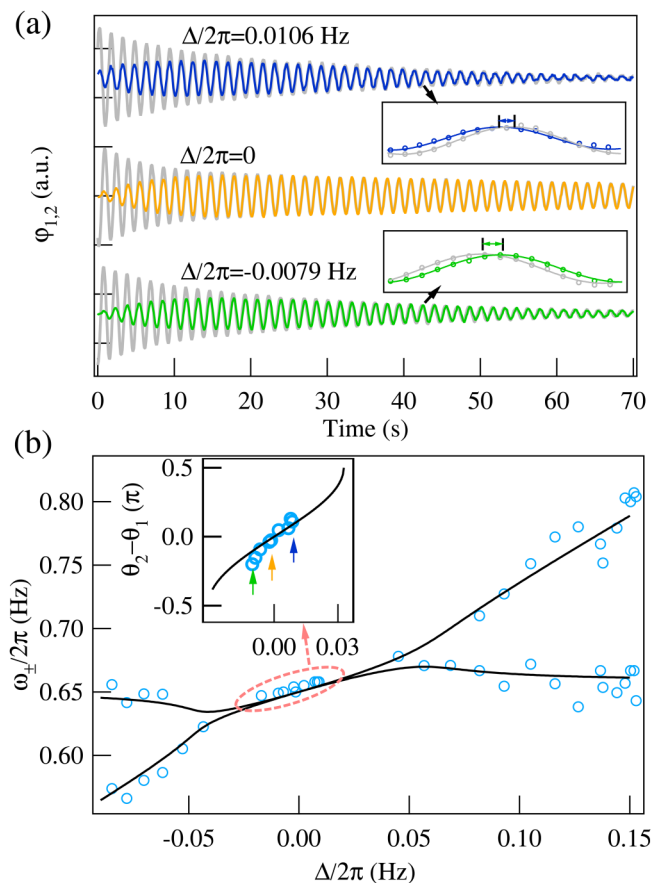


FIG. 6. (a) The comparison of $\varphi_1(t)$ and $\varphi_2(t)$ at $\Delta/2\pi = 0.0106$ Hz, $\Delta/2\pi = 0$, and $\Delta/2\pi = -0.0079$ Hz. The phase shifts between two pendulums are highlighted in the inset figures. The gray (color) curves correspond to the first (second) oscillator. Symbols (Solid lines) are experimental (simulation) results. (b) Oscillation frequencies (symbols) deduced from the FFT spectra in Fig. 5(b). Solid lines are calculated eigenfrequencies according to Eq. (4). The inset highlights the deduced phase difference (symbols) between two pendulums as a function of detuning inside the synchronization area, where the solid line follows a simple relation $\theta_2 - \theta_1 = \arcsin(\Delta/2\Gamma_1)$.

22 June 2026 03:10:54

to fit the experimental data (symbols) after a short period of time. The deduced phase shift $\theta_2 - \theta_1$ as a function of Δ is plotted as symbols in the inset of Fig. 6(b), which roughly follows a simple relation of $\theta_2 - \theta_1 = \arcsin(\Delta/2\Gamma_1)$ (solid line), in agreement with our previous discussion in Sec. II.

The symbols in Fig. 6(b) are the oscillation components obtained from the FFT spectra, which can be described by the solid line calculated according to Eq. (4). It should be noticed that the dispersions of $\omega_{\pm}$ vs Δ are not very symmetrical because Γ_2 increases rapidly with ω_2 . We first estimate Γ_2 (as a function of ω_2) according to $\Gamma_2 \propto L_2^{-2} \propto \omega_2^4$ using $\Gamma_1/2\pi = \Gamma_2/2\pi = 0.027$ Hz at $\Delta = 0$. With these determined parameters, we can calculate the frequencies of the hybridized modes based on Eq. (4), which is plotted as the solid lines in Fig. 6(b).

IV. SYNCHRONIZATION BASED ON DISSIPATIVELY COUPLED RLC CIRCUITS

After examining the synchronization in a dissipatively coupled mechanical system (with a characteristic resonance frequency in Hz), we now reference a study on MHz electrical oscillators, with a specific emphasis and analysis on synchronization resulting from dissipative coupling. For more comprehensive information, refer Ref. 52. This electrical circuit (Fig. 7) consists of two RLC resonators coupled to each other with a resistor. Different from previous studies in resistive coupling,^{53,54} the circuit is a purely linear system without any passive or active nonlinear elements. The two oscillators are coupled through the current $I_C = (V_2 - V_1)/R_C$ flowing through the resistor R_C . Defining $V_{1,2}$ as the variable and applying Kirchhoff's circuit laws that the sum of currents at any node of the circuit is zero, one obtains (discussed in more detail in Appendix C)

$$\text{coupling term} = 2\Gamma_{1,2}(\dot{V}_{2,1} - \dot{V}_{1,2}), \quad (14)$$

with the coupling coefficient $\Gamma_{1,2} = \frac{1}{2R_C C_{1,2}}$. In this situation, when the oscillator frequency $\omega_2 = \frac{1}{\sqrt{L_2 C_2}}$ is adjusted by the inductance L_2 , the coupling coefficient can still be maintained as constant ($\Gamma_1 = \Gamma_2 = \Gamma$) by setting $C_1 = C_2$. Appendix C gives the complete coupled differential equations of the circuit, and Appendix D provides the derivation process of its transmission coefficient S_{21} based on the ABCD matrix.⁵⁵

Assemble a coupled circuit with constant $C_1 = C_2$, constant L_1 , variable L_2 , and constant R_C . The parasitic resistance R_1 and R_2 are equal to each other by carefully designing the RLC circuit. These circuit parameters yield $\omega_1/2\pi = 9.31$ MHz, $\omega_2 = 7.52 - 11.26$ MHz, $\gamma_1/2\pi = \gamma_2/2\pi = 0.10$ MHz, and $\Gamma/2\pi = 0.25$ MHz. Therefore, a coupled RLC system with anti- $\mathcal{PT}$ symmetry is constructed.

The transmission coefficient S_{21} of the coupled RLC circuit measured by a vector network analyzer (VNA) is mapped as a function of frequency detunings $\omega - \omega_1$ and Δ in Fig. 8(a), where the characteristic level attraction (guided by dotted lines) with a coalescence of hybridized modes is clearly seen. Typical spectra in Figs. 8(b) and 8(c) further reveal that under matching conditions, the amplitude of S_{21} is an order of magnitude greater than the mode split case, and the linewidth is much narrower than the mode split case. In the same experimental setup, an oscilloscope (OSC) is used to measure the transient response of electrical pulses.

Then, the FFT is used to convert the OSC measurement value in the time-domain into a representation in the frequency-domain. The results with characteristics similar to the VNA measurements are summarized in the mapping shown in Fig. 8(d), and the typical FFT spectra are plotted in Figs. 8(e) and 8(f).

The measured frequency and linewidth of the hybridized mode are plotted in Figs. 8(g) and 8(h), respectively, which clearly show the characteristics of the anti- $\mathcal{PT}$ symmetric system. Here, the solid triangles and open circles are from the VNA and OSC measurements, respectively, and the solid curves are calculated according to Eq. (4). The dispersion again highlights the mode

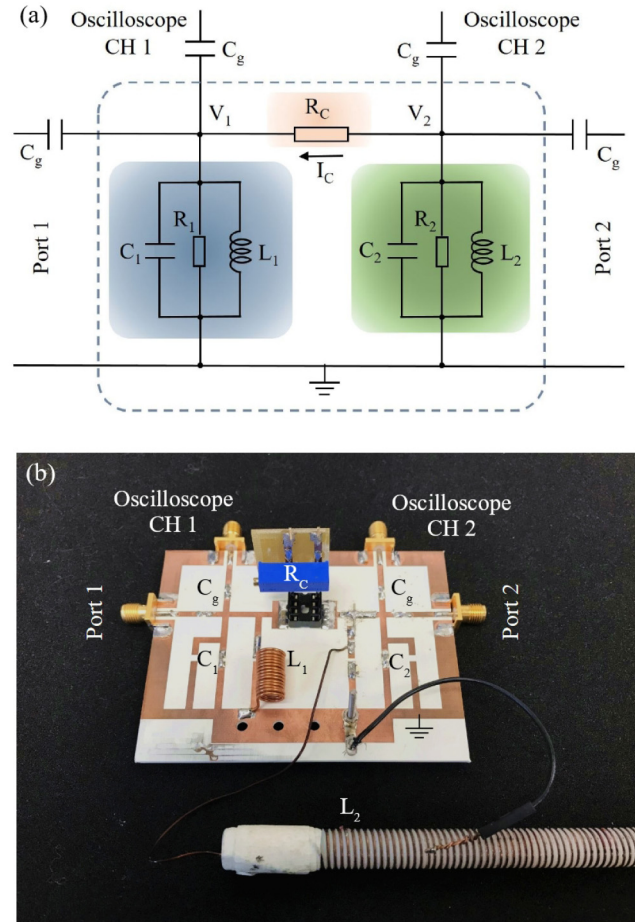


FIG. 7. (a) Circuit diagram and (b) picture of the actual electrical device of two dissipatively coupled RLC circuits. The mutual coupling is achieved by the electric current I_C through the variable resistor R_C . $L_{1,2}$, $C_{1,2}$, and $R_{1,2}$ denote the inductance, capacitance, and parasitic resistance. The voltage signals $V_1(t)$ and $V_2(t)$ can be recorded simultaneously by an oscilloscope. The S-parameter between port 1 and port 2 can be measured using a VNA. C_g is the coupling capacitance to isolate the measurement system and the circuit under test. Reproduced with permission from Kim *et al.*, AIP Adv. **12**, 035217 (2022). Copyright 2022 Author(s), licensed under a Creative Commons Attribution (CC BY) License.

coalescence. The evolution of the linewidth shows another characteristic of dissipative coupling, that is, the linewidth becomes narrower in a strongly coupled state. This feature has already shown in Figs. 8(c) and 8(f) but becomes more striking in Fig. 8(h). Near

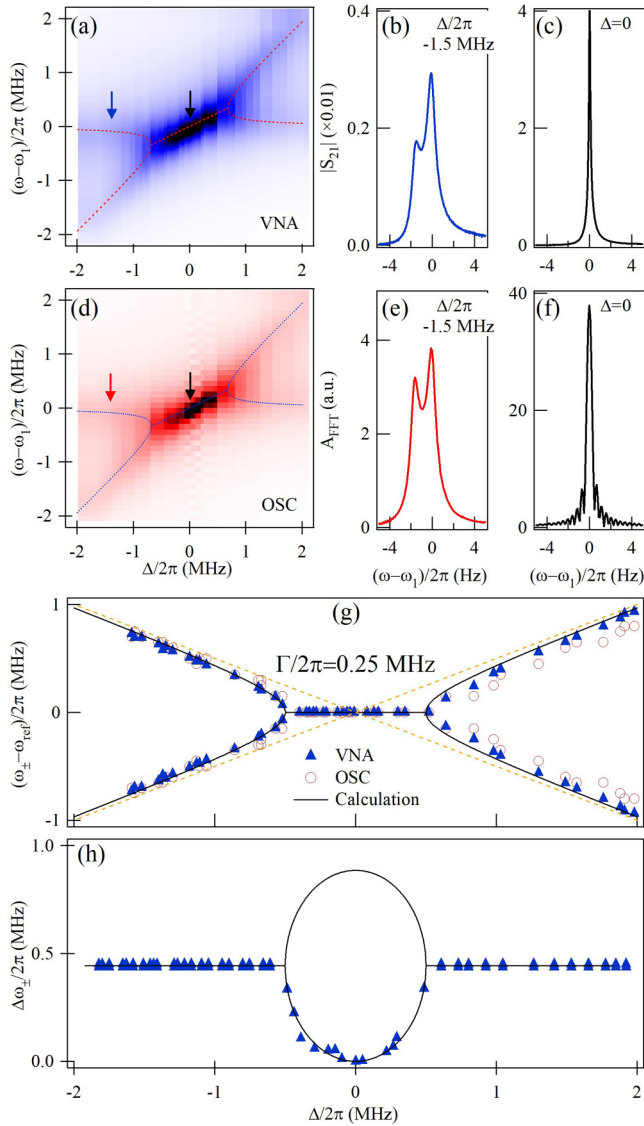


FIG. 8. (a) Transmission spectra S_{21} measured by a vector network analyzer (VNA) are mapped as a function of $\omega - \omega_1$ and Δ . (b) and (c) are the typical spectra at $\Delta/2\pi = -1.5$ MHz, and $\Delta=0$ [indicated by arrows in (a)], respectively. (d) FFT spectra measured by an oscilloscope (OSC) are mapped as a function of $\omega - \omega_1$ and Δ ; (e) and (f) are typical spectra. (g) and (h), respectively, plot the deduced resonance frequency and linewidth, where the solid curves are the calculated results using $\Gamma/2\pi = 0.25$ MHz. The diagonal dotted lines indicate the uncoupled oscillator frequency. Panels (g) and (h) incorporated VNA measurement results with permission from Kim *et al.*, AIP Adv. **12**, 035217 (2022). Copyright 2022 Author(s), licensed under a Creative Commons Attribution (CC BY) License.

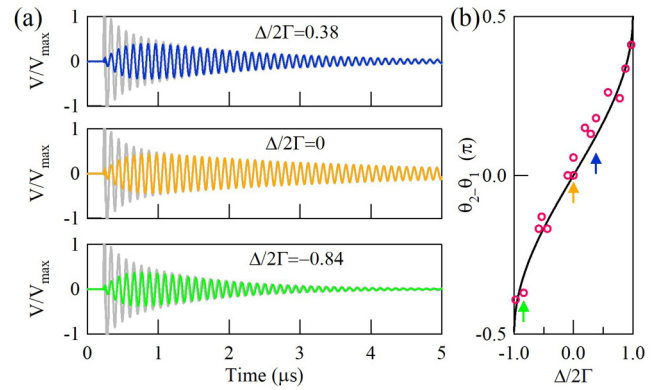


FIG. 9. (a) Typical time-domain measurement at different Δ . The gray (color) curves correspond to the first (second) oscillator. (b) Phase $\Theta = \theta_2 - \theta_1$ (symbols) vs detuning Δ follows a simple relation $\Theta = \arcsin(\Delta/2\Gamma)$.

$\Delta = 0$, the response of the heavily damped mode (anti-phase mode) is superimposed on the in-phase mode in the FFT spectra, and the latter has a much narrower linewidth and a greater amplitude, so it is difficult to precisely determine the linewidth of the anti-phase mode.

The time evolution of V_1 and V_2 are simultaneously recorded by using an OSC. As shown in Fig. 9(a), the characteristic synchronization pattern is clearly revealed inside the regime $(-2\Gamma < \Delta < 2\Gamma)$. During the first few cycles, the amplitude of the second oscillator increases rapidly, and its phase tends to align with the first oscillator. According to Eqs. (9a) and (9b), such a dynamic process can be understood, in which the anti-phase component is quickly quenched, leaving only the in-phase component. From these data, the phase difference between the oscillators is derived and summarized in Fig. 9(b) as symbols. As expected, the Θ - Δ relation follows a simple relation $\Theta = \arcsin(\Delta/2\Gamma)$.

One can also assemble a coupled circuit with anti- $\mathcal{PT}$ symmetry by varying the coupling strength. To do so, one fix the parameters of the second oscillator with $\omega_2/2\pi = 7.25$ MHz but vary the coupling resistance so that the coupling strength can be varied in a wide range $\Gamma/2\pi = 0.30 - 16.9$ MHz. The time-evolution of the second oscillator measured by the OSC and their FFT spectra are plotted in Figs. 10(a)-10(c). When $\Gamma \gg \Delta$ [Fig. 10(a)], the two oscillators tend to in-phase motion, and the net dissipation caused by the coupling effect is small, so that the signal slowly decays. Decreasing the coupling strength to $\Gamma \sim \Delta/2$ [Fig. 10(b)], the oscillators are still phase-locked so that its FFT spectrum only shows one broader peak. Further decreasing the coupling strength to $\Gamma \ll \Delta/2$ [Fig. 10(c)], a beating pattern is clearly seen because of the interference between two hybridized modes with slightly different frequencies. The deduced resonance frequencies from the FFT as well as the results measured by a VNA are summarized in Fig. 10(d), where the phase broken of the anti- $\mathcal{PT}$ symmetry is demonstrated with a threshold value of $\Delta = -2\Gamma$. Again, in the exact anti- $\mathcal{PT}$ -symmetry phase, the two synchronized oscillators have a phase difference following a simple

22 June 2026 03:10:54

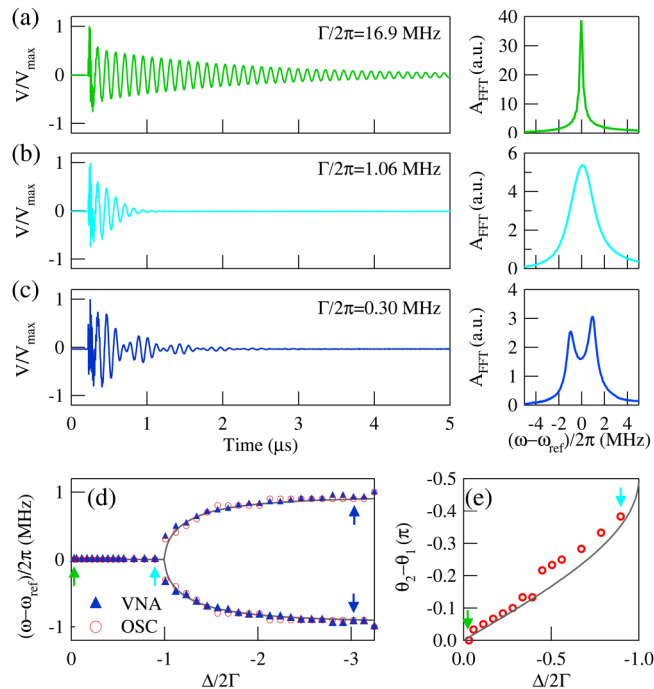


FIG. 10. (a)–(c) Typical time-domain spectra and their FFT spectra at different coupling strengths by fixing $\Delta/2\pi = -2.06$ MHz. (d) Anti- $\mathcal{PT}$ -symmetry phase broken vs Δ/Γ . (e) Phase difference between two oscillators as a function of Δ/Γ inside the synchronization area, where the solid line is calculated according to an $\arcsin(\Delta/2\Gamma)$ relation. Panel (d) incorporated VNA measurement results with permission from Kim *et al.*, AIP Adv. **12**, 035217 (2022). Copyright 2022 Author(s), licensed under a Creative Commons Attribution (CC BY) License.

$\arcsin(\Delta/2\Gamma)$ relation, as shown in Fig. 10(e). This circuit can be also used as an education tool to visualize eigenvectors and synchronization zone on the Bloch sphere.⁵²

V. DISCUSSIONS, CONCLUSIONS, AND PERSPECTIVE

The history of synchronization can find its roots in the discovery made by Christiaan Huygens in 1665. While conducting experiments with pendulum clocks, Huygens observed that when he placed two of these timekeeping devices side by side on a common support, they often exhibited a remarkable tendency to synchronize. Over the next nearly 360 years, researchers have continued to delve into this fascinating phenomenon, revisiting it time and again. New perspectives, topics, implementations, and applications have transformed synchronization from a mere curiosity into a dynamic and multifaceted field at the intersection of science, mathematics, and technology. This Tutorial outlines the basics of synchronization caused by dissipative coupling, where the dissipative dynamics result in the phase-locking of two non-identical oscillators. Building upon the intriguing findings observed in a spring-coupled pendulum system, which ignites students' curiosity and motivates them to explore intricate realms of classical mechanics

and advanced quantum mechanics, our goal is to guide the next generation of researchers in their understanding of synchronization dynamics through a simplified model. We present two distinct dissipative coupled linear systems—specifically, two pendulums coupled through the back EMF induced by the Lenz effect and two RLC circuits interconnected through resistive coupling. These educational tools are tailored for advanced graduate-level studies.

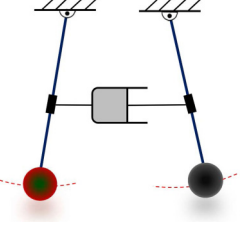
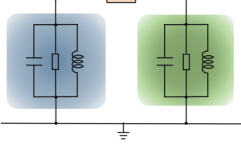
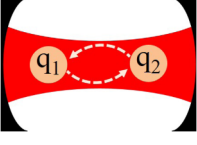
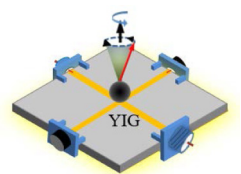
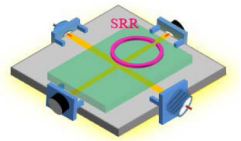
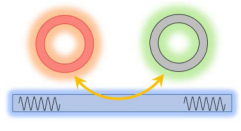
In recent years, dissipative coupling has been successfully demonstrated across a range of advanced systems, including microwave optomechanical circuits,⁶⁸ coupled cavity-magnon systems,^{37,38} coupled metamaterial structures,⁶³ and coupled mechanical-atomic spin systems.⁶⁹ This intriguing phenomenon manifests as level attraction, leading to a coalescence of hybridized modes. The observation sharply contrasts with the level repulsion typically observed in coherent coupling scenarios, which leads to mode anticrossing. The distinct advantage of employing dissipative coupling is its capacity to clearly showcase synchronization dynamics to a broad, no-expert audience, entirely avoiding the need for complex nonlinear mathematics.

Interestingly, the dissipative coupling can produce a coupled system with the anti- $\mathcal{PT}$ symmetry, i.e., the eigenfrequency splitting for the broken anti- $\mathcal{PT}$ symmetry phase and linewidth splitting for the exact anti- $\mathcal{PT}$ symmetry phase. Here, the transition condition is the EP of the anti- $\mathcal{PT}$ symmetry. Synchronization only occurs in the anti- $\mathcal{PT}$ -symmetry phase. The phase difference between the two oscillators that appears in the time domain measurement is related to eigenstate, determined by the eigenvector of the lower damping state.

In addition to these tutorial models designed to elucidate the fundamentals of dissipatively coupled systems and synchronization, researchers can delve into these concepts, along with other intriguing physics phenomena within coupled oscillator systems covering different frequency ranges, as outlined in Table II. For example, recent research studies have revealed that dissipative coupling also supports a bound state in the continuum (BIC).^{40,63} BICs, also known as embedded trapped modes, correspond to discrete eigenvalues coexisting with extended modes of a continuous spectrum.^{70,71}

Strictly speaking, the third system and the last three systems in Table II are base-mediated coupling, similar to the famous example of Huygens' clocks, in which two pendulums are coupled through a common mounting support. This base-mediated system uses the common base as a mutual reservoir, allowing energy leaked by one oscillator to drive the second indirectly. Depending on the phase correlation between fields emanated by the two oscillators, the indirect coupling can be coherent or dissipative or a combination of them.³¹ Interestingly, the strength of the dissipative coupling may be negative (corresponds to “negative friction,” a force that promotes relative movement). As a result, in addition to the in-phase synchronization described in the Kuramoto model, the anti-phase synchronization phenomenon that Huygens first observed can be explained by setting Γ to a negative number in Eq. (8). Experimentally, the dissipatively coupled pendulum system described in this Tutorial can be used to demonstrate anti-phase synchronization by manipulating the coupling strength and type via a feedback architecture. This exhibition was achieved by introducing a velocity-dependent-current into the coil to generate

TABLE II. Interesting physics within diverse coupled systems for future tutorial study. LA, level attraction; Sync, synchronization; NR, nonreciprocity; SP, spin pumping; and BIC, bound states in the continuum.

Device and setup	Parameters	Experimental observation
	Mechanical system $\omega/2\pi \sim 1$ Hz $\gamma/2\pi \sim 1$ mHz $\Gamma/2\pi \sim 0.1$ Hz	Tutorial model - LA - Sync
	Electronics $\omega/2\pi \sim 1$ kHz–10 MHz $\gamma/2\pi \sim 10^{-2} - 10^2$ kHz $\Gamma/2\pi \sim 0.1 - 10^3$ kHz	Tutorial model - LA - Sync - anti-$\mathcal{PT}$^{52,53} - Chaos⁵⁴
	Phononics $\omega/2\pi \sim 1$ MHz $\gamma/2\pi \sim 1$ Hz $\Gamma/2\pi \sim \text{N/A}$	Sync^{56–58}
	Magnonics $\omega/2\pi \sim 1$ GHz $\gamma/2\pi \sim 1 - 10$ MHz $\Gamma/2\pi \sim 10^{-2} - 0.1$ GHz	- LA³⁴ - SP^{33–35,59}
	Magnon polariton $\omega/2\pi \sim 1$ GHz $\gamma/2\pi \sim 1 - 10$ MHz $\Gamma/2\pi \sim 10^{-2} - 0.1$ GHz	- LA^{37,38} - Sync⁶⁰ - BIC⁴⁰ - NR⁶¹ - anti-$\mathcal{PT}$^{40,62}
	Metamaterials $\omega/2\pi \sim 1$ GHz–1 THz $\gamma/2\pi \sim 10 - 10^4$ MHz $\Gamma/2\pi \sim 0.1 - 10^2$ GHz	- LA⁶³ - Sync⁶⁴ - BIC⁶³
	Optics and lasers $\omega/2\pi \sim 30$ THz $\gamma/2\pi$ N/A $\Gamma/2\pi$ N/A	Sync^{65–67}

Ampere's force between the coil and magnet, as comprehensively explained in Ref. 51.

In summary, the technology of constructing and engineering the dissipative coupling between oscillators represents a promising platform in studying synchronization, singularity, eigenvectors in

hybridized systems, topological phase transition, and wave manipulation.

Note added in proof. Recently, a preprint by Moreno *et al.*⁷² appeared, exploring the synchronization within dissipatively coupled harmonic oscillator networks.

ACKNOWLEDGMENTS

This work was funded by NSERC Discovery Grants and NSERC Discovery Accelerator Supplements (C.-M.H.). We thank Yutong Zhao, Yipu Wang, Jinwei Rao, Jie Qian, Jaehyeong Yang, Jacob Burgess, William E. Bailey, Mohammad-Ali Miri, Girish Saran Agarwal, Hong Chen, Ke Xia, Jiang Xiao, Yang Xiao, and Andreas Nunnenkamp for discussions.

AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Author Contributions

Chenyang Lu: Conceptualization (equal); Formal analysis (equal); Investigation (equal); Validation (equal). **Mun Kim:** Conceptualization (equal); Formal analysis (equal); Investigation (equal); Validation (equal). **Ying Yang:** Formal analysis (equal); Validation (equal). **Y. S. Gui:** Conceptualization (equal); Formal analysis (equal); Methodology (equal); Writing – original draft (equal); Writing – review & editing (equal). **C.-M. Hu:** Conceptualization (equal); Supervision (lead); Writing – original draft (equal); Writing – review & editing (equal).

DATA AVAILABILITY

The data that support the findings of this study are available from the corresponding author upon reasonable request.

APPENDIX A: METHOD OF AVERAGING

In this Appendix, we provide some details for analyzing the time-evolution of dynamic processes using the method of averaging. We focus solely on pure dissipative coupling ($J_1 = J_2 = 0$) and begin by rewriting the equations of motion (2) as

$$\ddot{x}_{1,2} + \omega_{ref}^2 x_{1,2} + h_{1,2} = 0. \quad (A1)$$

$h_{1,2} = 2\gamma_{1,2}\dot{x}_{1,2} + (\omega_{1,2}^2 - \omega_{ref}^2)x_{1,2} - 2\Gamma_{1,2}(\dot{x}_{2,1} - \dot{x}_{1,2})$ are small perturbations. The solution of the stationary states has a form as $x_{1,2} = A_{1,2}(t)\cos[\omega_{ref}t + \theta_{1,2}(t)]$, which contains time scales separation: a fast oscillation $\cos(\omega_{ref}t)$ vs a slow drift in the amplitude $A_{1,2}(\geq 0)$ and phase $\theta_{1,2}$. Due to their slowly varying nature, $A_{1,2}$ and $\theta_{1,2}$ can be determined using the method of averages,^{4,28}

$$\frac{dA_{1,2}}{dt} = \langle h_{1,2}\sin(\tau) \rangle, \quad (A2a)$$

$$A_{1,2}\frac{d\theta_{1,2}}{dt} = \langle h_{1,2}\cos(\tau) \rangle, \quad (A2b)$$

where $\langle * \rangle$ denotes a time average over the period $T = 2\pi/\omega_{ref}$ and $\tau = \omega_{ref}t + \theta_{1,2}$ for $h_{1,2}$, respectively. Since $A_{1,2}$ and $\theta_{1,2}$ are approximately constant over T , time averaging $h_{1,2}$ leads to

$$\frac{dA_1}{dt} = -(\gamma_1 + \Gamma_1)A_1 + \Gamma_1A_2\cos(\theta_2 - \theta_1), \quad (A3a)$$

$$\frac{dA_2}{dt} = -(\gamma_2 + \Gamma_2)A_2 + \Gamma_2A_1\cos(\theta_2 - \theta_1), \quad (A3b)$$

$$\frac{d(\theta_2 - \theta_1)}{dt} = \Delta - \left(\frac{A_1\Gamma_2}{A_2} + \frac{A_2\Gamma_1}{A_1} \right) \sin(\theta_2 - \theta_1). \quad (A3c)$$

It is worth noting that the coupling term only depends on the phase difference as described in the Kuramoto model. In the absence of coupling, the amplitude of each oscillator would simply decay exponentially with a time constant determined by its own damping, and their phase difference would advance at a constant rate.

APPENDIX B: EQUATIONS OF MOTION FOR DISSIPATIVE COUPLED PENDULUMS

Referring to dissipative coupled pendulums (Fig. 4), for small amplitudes, $\sin(\phi_{1,2}) \approx \phi_{1,2}$ while $\cos(\phi_{1,2}) \approx 1 - \phi_{1,2}^2/2$, and therefore, the Lagrangian for this coupled pendulum system is

$$\mathcal{L} = T - U = \frac{1}{2}m\dot{\phi}_1^2 + \frac{1}{2}m\dot{\phi}_2^2 - \frac{1}{2}mgl_1\phi_1^2 - \frac{1}{2}mgl_2\phi_2^2, \quad (B1)$$

where T and U are the kinetic energy and the total potential energy. Different from spring-induced coherent coupling, there is no potential energy associated with dissipative coupling. Instead, the Lenz effect introduces a velocity proportional force in Rayleigh's dissipation function,

$$\mathcal{F} = \gamma_1 m \dot{\phi}_1^2 + \gamma_2 m \dot{\phi}_2^2 + \nu m (\dot{\phi}_1 - \dot{\phi}_2)^2, \quad (B2)$$

where $\gamma_{1,2}$ characterizes the intrinsic damping rate of the pendulums and the kinematic viscosity ν acts as a proportionality constant, characterizing the back EMF ($\mathcal{E} = -d\Phi_B/dt$) between the two pendulums caused by the Lenz effect. The Lenz's law states that the induced EMF $\mathcal{E}$ and the rate of change in magnetic flux Φ_B has opposite signs. In our setup, it is clear that the rate of change in magnetic flux is proportional to the relative velocity between two pendulums.

Therefore, the equations of motion for $\phi_{1,2}$ can be derived according to the generalized Euler-Lagrange equations,

$$\frac{d}{dt} \frac{\partial \mathcal{L}}{\partial \dot{\phi}_i} - \frac{\partial \mathcal{L}}{\partial \phi_i} + \frac{\partial \mathcal{F}}{\partial \dot{\phi}_i} = 0, \quad i = 1, 2, \quad (B3)$$

to be

$$\ddot{\phi}_1 + 2\gamma_1\dot{\phi}_1 + \omega_1^2\phi_1 = 2\Gamma_1(\dot{\phi}_2 - \dot{\phi}_1), \quad (B4a)$$

$$\ddot{\phi}_2 + 2\gamma_2\dot{\phi}_2 + \omega_2^2\phi_2 = 2\Gamma_2(\dot{\phi}_1 - \dot{\phi}_2), \quad (B4b)$$

where $\omega_{1,2} = \sqrt{g/l_{1,2}}$ is the uncoupled oscillation frequency and $\Gamma_{1,2} = \nu/l_{1,2}^2 \ll \omega_{1,2}$ describes the coupling strength. Obviously, Eqs. (B4) and (2) (at $f_1 = f_2 = 0$ and $J_1 = J_2 = 0$) are mathematically equivalent. Taking the approach as in Sec. II, we can directly

compare the measurement results of coupled pendulums with our model.

APPENDIX C: EQUATIONS OF MOTION FOR DISSIPATIVE COUPLED RLC CIRCUITS

Applying Kirchhoff's circuit laws for the proposed electric circuit in Fig. 7, we obtain

$$\frac{V_1}{R_1} + \int \frac{V_1}{L_1} dt + C_1 \dot{V}_1 = I_C, \quad (\text{C1a})$$

$$(V_2 - V_1)/R_C = I_C, \quad (\text{C1b})$$

$$\frac{V_2}{R_2} + \int \frac{V_2}{L_2} dt + C_2 \dot{V}_2 = -I_C. \quad (\text{C1c})$$

Substituting Eq. (C1b) into Eqs. (C1a) and (C1c) leads to coupled differential equations

$$\ddot{V}_1 + \frac{1}{C_1 R_1} \dot{V}_1 + \frac{V_1}{L_1 C_1} = \frac{1}{R_C C_1} (\dot{V}_2 - \dot{V}_1), \quad (\text{C2a})$$

$$\ddot{V}_2 + \frac{1}{C_2 R_2} \dot{V}_2 + \frac{V_2}{L_2 C_2} = \frac{1}{R_C C_2} (\dot{V}_1 - \dot{V}_2). \quad (\text{C2b})$$

Defining $\omega_{1,2} = \frac{1}{\sqrt{L_{1,2} C_{1,2}}}$, $\gamma_{1,2} = \frac{1}{2C_{1,2} R_{1,2}}$, and $\Gamma_{1,2} = \frac{1}{2C_{1,2} R_C}$, it can be found that Eq. (C2) for our coupled RLC circuits is identical as Eq. (B4) for dissipatively coupled pendulums.

APPENDIX D: SCATTERING PARAMETERS OF DISSIPATIVE COUPLED RLC CIRCUITS

In this Appendix, we present some details of the analytical solution for scattering parameters (S-parameters) of dissipative coupled RLC circuits. We start with the transfer matrix (the ABCD matrix in microwave terminology) following the $e^{j\omega t}$ convention. For a cascaded system of many components, such as in Fig. 7, the transfer matrix between ports 1 and 2 is determined by multiplying the ABCD matrix for each component as⁵⁵

$$M = \begin{pmatrix} A & B \\ C & D \end{pmatrix} = M_g M_1 M_C M_2 M_g. \quad (\text{D1})$$

The transfer matrix M_g of the capacitance coupling section is

$$M_g = \begin{pmatrix} 1 & \frac{1}{j\omega C_g} \\ 0 & 1 \end{pmatrix}. \quad (\text{D2})$$

The transfer matrix $M_{1,2}$ of the oscillator section is

$$M_{1,2} = \begin{pmatrix} 1 & 0 \\ Y_{1,2} & 1 \end{pmatrix}, \quad (\text{D3})$$

where $Y_{1,2} = 1/R_{1,2} + 1/(j\omega L_{1,2}) + j\omega C_{1,2}$ is the admittance of the RLC circuit.

The transfer matrix M_C of the coupling section is

$$M_C = \begin{pmatrix} 1 & R_C \\ 0 & 1 \end{pmatrix}. \quad (\text{D4})$$

Then, the S-parameter between ports 1 and 2 can be converted from the ABCD matrix according to Ref. 55,

$$\begin{pmatrix} S_{11} & S_{12} \\ S_{21} & S_{22} \end{pmatrix} = \begin{pmatrix} \frac{A+B/Z_0-CZ_0-D}{A+B/Z_0+CZ_0+D} & \frac{2(AD-BC)}{A+B/Z_0+CZ_0+D} \\ \frac{2}{A+B/Z_0+CZ_0+D} & \frac{-A+B/Z_0-CZ_0+D}{A+B/Z_0+CZ_0+D} \end{pmatrix}. \quad (\text{D5})$$

Z_0 is the characteristic impedance of the transmission line.

Since the coupling capacitance is very small [i.e., $Z_0, R_C \ll 1/(j\omega C_g)$], we can obtain a simple expression of the microwave transmission as

$$S_{21}(\omega) = -\frac{2Z_0 R_C \omega^2 C_g^2 \Gamma_1 \Gamma_2}{[j(\omega - \omega_1) + \gamma_1 + \Gamma_1][j(\omega - \omega_2) + \gamma_2 + \Gamma_2] - \Gamma_1 \Gamma_2}. \quad (\text{D6})$$

Here, $\omega_{1,2} = \frac{1}{\sqrt{L_{1,2} C_{1,2}}}$, $\gamma_{1,2} = \frac{1}{2C_{1,2} R_{1,2}}$, and $\Gamma_{1,2} = \frac{1}{2C_{1,2} R_C}$ are the same as defined in Appendix C.

REFERENCES

- ¹L. M. Pecora and T. L. Carroll, "Synchronization in chaotic systems," *Phys. Rev. Lett.* **64**, 821 (1990).
- ²S. H. Strogatz, "From Kuramoto to Crawford: Exploring the onset of synchronization in populations of coupled oscillators," *Physica D* **143**, 1–20 (2000).
- ³A. T. Winfree, *The Geometry of Biological Time* (Springer Science & Business Media, 2001), Vol. 12.
- ⁴A. Pikovsky, M. Rosenblum, and J. Kurths, *Synchronization: A Universal Concept in Nonlinear Sciences* (Cambridge University Press, 2001), Vol. 12.
- ⁵S. Strogatz, *Sync: The Emerging Science of Spontaneous Order* (Penguin UK, 2004).
- ⁶A. T. Winfree, "Biological rhythms and the behavior of populations of coupled oscillators," *J. Theor. Biol.* **16**, 15–42 (1967).
- ⁷S. M. Armstrong, V. M. Cassone, M. J. Chesworth, J. R. Redman, and R. V. Short, "Synchronization of mammalian circadian rhythms by melatonin," *J. Neural Transm. Supplementum* **21**, 375–394 (1986).
- ⁸S. J. Aton and E. D. Herzog, "Come together, right...now: Synchronization of rhythms in a mammalian circadian clock," *Neuron* **48**, 531–534 (2005).
- ⁹C. Schäfer, M. G. Rosenblum, J. Kurths, and H.-H. Abel, "Heartbeat synchronized with ventilation," *Nature* **392**, 239–240 (1998).
- ¹⁰J. H. McAuley and C. D. Marsden, "Physiological and pathological tremors and rhythmic central motor control," *Brain* **123**, 1545–1567 (2000).
- ¹¹L. Glass, "Synchronization and rhythmic processes in physiology," *Nature* **410**, 277–284 (2001).
- ¹²B. D. Josephson, "Coupled superconductors," *Rev. Mod. Phys.* **36**, 216 (1964).
- ¹³K. Wiesenfeld, P. Colet, and S. H. Strogatz, "Synchronization transitions in a disordered Josephson series array," *Phys. Rev. Lett.* **76**, 404 (1996).
- ¹⁴P. Barbara, A. B. Cawthorne, S. V. Shitov, and C. J. Lobb, "Stimulated emission and amplification in Josephson junction arrays," *Phys. Rev. Lett.* **82**, 1963 (1999).
- ¹⁵S. Kaka, M. R. Pufall, W. H. Rippard, T. J. Silva, S. E. Russek, and J. A. Katine, "Mutual phase-locking of microwave spin torque nanooscillators," *Nature* **437**, 389–392 (2005).

- ¹⁶F. B. Mancoff, N. D. Rizzo, B. N. Engel, and S. Tehrani, "Phaselocking in double-point-contact spin-transfer devices," *Nature* **437**, 393–395 (2005).
- ¹⁷A. A. Awad, P. Dürrenfeld, A. Houshang, M. Dvornik, E. Iacocca, R. K. Dumas, and J. Åkerman, "Long-range mutual synchronization of spin Hall nano-oscillators," *Nat. Phys.* **13**, 292–299 (2017).
- ¹⁸S.-B. Shim, M. Imboden, and P. Mohanty, "Synchronized oscillation in coupled nanomechanical oscillators," *Science* **316**, 95–99 (2007).
- ¹⁹M. Zhang, G. S. Wiederhecker, S. Manipatruni, A. Barnard, P. McEuen, and M. Lipson, "Synchronization of micromechanical oscillators using light," *Phys. Rev. Lett.* **109**, 233906 (2012).
- ²⁰M. Aspelmeyer, T. J. Kippenberg, and F. Marquardt, "Cavity optomechanics," *Rev. Mod. Phys.* **86**, 1391 (2014).
- ²¹M. Rohden, A. Sorge, M. Timme, and D. Witthaut, "Self-organized synchronization in decentralized power grids," *Phys. Rev. Lett.* **109**, 064101 (2012).
- ²²A. E. Motter, S. A. Myers, M. Anghelescu, and T. Nishikawa, "Spontaneous synchrony in power-grid networks," *Nat. Phys.* **9**, 191–197 (2013).
- ²³L. M. Pecora, F. Sorrentino, A. M. Hagerstrom, T. E. Murphy, and R. Roy, "Cluster synchronization and isolated desynchronization in complex networks with symmetries," *Nat. Commun.* **5**, 4079 (2014).
- ²⁴A. Warsi Laskar, P. Adhikary, S. Mondal, P. Katiyar, S. Vinjanampathy, and S. Ghosh, "Observation of quantum phase synchronization in spin-1 atoms," *Phys. Rev. Lett.* **125**, 013601 (2020).
- ²⁵H. G. Winful and L. Rahman, "Synchronized chaos and spatiotemporal chaos in arrays of coupled lasers," *Phys. Rev. Lett.* **65**, 1575 (1990).
- ²⁶D. J. DeShazer, R. Breban, E. Ott, and R. Roy, "Detecting phase synchronization in a chaotic laser array," *Phys. Rev. Lett.* **87**, 044101 (2001).
- ²⁷J. Ding, I. Belykh, A. Marandi, and M. Ali Miri, "Dispersive versus dissipative coupling for frequency synchronization in lasers," *Phys. Rev. Appl.* **12**, 054039 (2019).
- ²⁸S. H. Strogatz, *Nonlinear Dynamics and Chaos with Student Solutions Manual: With Applications to Physics, Biology, Chemistry, and Engineering* (CRC Press, 2018).
- ²⁹D. G. Aronson, G. Bard Ermentrout, and N. Kopell, "Amplitude response of coupled oscillators," *Physica D* **41**, 403–449 (1990).
- ³⁰M. C. Cross, J. L. Rogers, R. Lifshitz, and A. Zumdick, "Synchronization by reactive coupling and nonlinear frequency pulling," *Phys. Rev. E* **73**, 036205 (2006).
- ³¹A. Metelmann and A. A. Clerk, "Nonreciprocal photon transmission and amplification via reservoir engineering," *Phys. Rev. X* **5**, 021025 (2015).
- ³²Y.-P. Wang and C.-M. Hu, "Dissipative couplings in cavity magnonics," *J. Appl. Phys.* **127**, 130901 (2020).
- ³³B. Heinrich, Y. Tserkovnyak, G. Woltersdorf, A. Brataas, R. Urban, and G. E. W. Bauer, "Dynamic exchange coupling in magnetic bilayers," *Phys. Rev. Lett.* **90**, 187601 (2003).
- ³⁴Y. Tserkovnyak, A. Brataas, G. E. W. Bauer, and B. I. Halperin, "Nonlocal magnetization dynamics in ferromagnetic heterostructures," *Rev. Mod. Phys.* **77**, 1375 (2005).
- ³⁵P. Hyde, L. Bai, D. M. J. Kumar, B. W. Southern, C.-M. Hu, S. Y. Huang, B. F. Miao, and C. L. Chien, "Electrical detection of direct and alternating spin current injected from a ferromagnetic insulator into a ferromagnetic metal," *Phys. Rev. B* **89**, 180404(R) (2014).
- ³⁶Y. Li, W. Cao, V. P. Amin, Z. Zhang, J. Gibbons, J. Sklenar, J. Pearson, P. M. Haney, M. D. Stiles, W. E. Bailey, V. Novosad, A. Hoffmann, and W. Zhang, "Coherent spin pumping in a strongly coupled magnon-magnon hybrid system," *Phys. Rev. Lett.* **124**, 117202 (2020).
- ³⁷M. Harder, Y. Yang, B. M. Yao, C. H. Yu, J. W. Rao, Y. S. Gui, R. L. Stamps, and C.-M. Hu, "Level attraction due to dissipative magnon-photon coupling," *Phys. Rev. Lett.* **121**, 137203 (2018).
- ³⁸B. Bhoi, B. Kim, S.-H. Jang, J. Kim, J. Yang, Y.-J. Cho, and S.-K. Kim, "Abnormal anticrossing effect in photon-magnon coupling," *Phys. Rev. B* **99**, 134426 (2019).
- ³⁹J. Okołowicz, M. Płoszajczak, and I. Rotter, "Dynamics of quantum systems embedded in a continuum," *Phys. Rep.* **374**, 271–383 (2003).
- ⁴⁰Y. Yang, Y.-P. Wang, J. W. Rao, Y. S. Gui, B. M. Yao, W. Lu, and C. M. Hu, "Unconventional singularity in anti-paritytime symmetric cavity magnonics," *Phys. Rev. Lett.* **125**, 147202 (2020).
- ⁴¹Y. Kuramoto, "Self-entrainment of a population of coupled nonlinear oscillators," in *International Symposium on Mathematical Problems in Theoretical Physics*, Lecture Notes in Physics Vol. 39, edited by H. Araki (Springer, Berlin, 1975), pp. 420–422.
- ⁴²Y. Kuramoto, "Cooperative dynamics of oscillator community: A study based on lattice of rings," *Prog. Theor. Phys. Supplement* **79**, 223–240 (1984).
- ⁴³F. A. Rodrigues, T. K. D. Peron, P. Ji, and J. Kurths, "The Kuramoto model in complex networks," *Phys. Rep.* **610**, 1 (2016).
- ⁴⁴M. C. Cross, A. Zumdick, R. Lifshitz, and J. L. Rogers, "Synchronization by nonlinear frequency pulling," *Phys. Rev. Lett.* **93**, 224101 (2004).
- ⁴⁵A. P. Kuznetsov, N. V. Stankevich, and L. V. Turukina, "Coupled van der Pol-Duffing oscillators: Phase dynamics and structure of synchronization tongues," *Physica D* **238**, 1203–1215 (2009).
- ⁴⁶W. D. Heiss, "The physics of exceptional points," *J. Phys. A: Math. Theor.* **45**, 444016 (2012).
- ⁴⁷M.-A. Miri and A. Alù, "Exceptional points in optics and photonics," *Science* **363**, eaar7709 (2019).
- ⁴⁸H. Zhang, R. Huang, S.-D. Zhang, Y. Li, C.-W. Qiu, F. Nori, and H. Jing, "Breaking anti-PT symmetry by spinning a resonator," *Nano Lett.* **20**, 7594 (2020).
- ⁴⁹J. M. P. Nair, D. Mukhopadhyay, and G. S. Agarwal, "Enhanced sensing of weak anharmonicities through coherences in dissipatively coupled anti-PT symmetric systems," *Phys. Rev. Lett.* **126**, 180401 (2021).
- ⁵⁰C. Lu, B. Turner, Y. Gui, J. Burgess, J. Xiao, and C.-M. Hu, "An experimental demonstration of level attraction with coupled pendulums," *Am. J. Phys.* **91**, 585 (2023).
- ⁵¹C. Lu, "Dissipative coupling in mechanical systems," Master's thesis (University of Manitoba, 2022).
- ⁵²M. Kim, Y. Yang, Y. S. Gui, and C.-M. Hu, "Visualization of synchronization zone on the Bloch sphere through an anti-PT-symmetric electrical circuit," *AIP Adv.* **12**, 035217 (2022).
- ⁵³Y. Choi, C. Hahn, J. Woong Yoon, and S. Ho Song, "Observation of an anti-PT-symmetric exceptional point and energy-difference conserving dynamics in electrical circuit resonators," *Nat. Commun.* **9**, 2182 (2018).
- ⁵⁴N. F. Rulkov, "Images of synchronized chaos: Experiments with circuits," *Chaos* **6**, 262–279 (1996).
- ⁵⁵D. M. Pozar, *Microwave Engineering* (John Wiley & Sons, 2009).
- ⁵⁶M. Bagheri, M. Poot, L. Fan, F. Marquardt, and H. X. Tang, "Photonic cavity synchronization of nanomechanical oscillators," *Phys. Rev. Lett.* **111**, 213902 (2013).
- ⁵⁷F. Bermani, A. Motazedifard, R. Roknizadeh, M. H. Naderi, and D. Vitali, "Synchronization dynamics of two nanomechanical membranes within a Fabry-Perot cavity," *Phys. Rev. A* **96**, 023805 (2017).
- ⁵⁸J. Sheng, X. Wei, C. Yang, and H. Wu, "Self-organized synchronization of phonon lasers," *Phys. Rev. Lett.* **124**, 053604 (2020).
- ⁵⁹H. Wang, J. Chen, T. Yu, C. Liu, C. Guo, H. Jia, S. Liu, K. Shen, T. Liu, J. Zhang, M. A. Cabero Z, Q. Song, S. Tu, M. Wu, X. Han, K. Xia, D. Yu, G. E. W. Bauer, and H. Yu, "Non-Hermitian coherent coupling of nanomagnets by exchange spin waves," *arXiv:2005.10452* (2020).
- ⁶⁰V. L. Grigoryan, K. Shen, and K. Xia, "Synchronized spin-photon coupling in a microwave cavity," *Phys. Rev. B* **98**, 024406 (2018).
- ⁶¹Y.-P. Wang, J. W. Rao, Y. Yang, P.-C. Xu, Y. S. Gui, B. M. Yao, J. Q. You, and C.-M. Hu, "Nonreciprocity and unidirectional invisibility in cavity magnonics," *Phys. Rev. Lett.* **123**, 127202 (2019).
- ⁶²J. Zhao, Y. Liu, L. Wu, C.-K. Duan, Y.-X. Liu, and J. Du, "Observation of anti-PT-symmetry phase transition in the magnon-cavity-magnon coupled system," *Phys. Rev. Appl.* **13**, 014053 (2020).
- ⁶³J. W. Rao, Y. T. Zhao, Y. S. Gui, X. L. Fan, D. S. Xue, and C.-M. Hu, "Controlling microwaves in non-hermitian metamaterials," *Phys. Rev. Appl.* **15**, L021003 (2021).
- ⁶⁴I. Al-Naib, E. Hebestreit, C. Rockstuhl, F. Lederer, D. Christodoulides, T. Ozaki, and R. Morandotti, "Conductive coupling of split ring resonators: A

path to THz metamaterials with ultrasharp resonances,” *Phys. Rev. Lett.* **112**, 183903 (2014).

⁶⁵L. Fabiny, P. Colet, R. Roy, and D. Lenstra, “Coherence and phase dynamics of spatially coupled solid-state lasers,” *Phys. Rev. A* **47**, 4287 (1993).

⁶⁶R. Roy and K. Scott Thornburg, Jr., “Experimental synchronization of chaotic lasers,” *Phys. Rev. Lett.* **72**, 2009 (1994).

⁶⁷T. Sugawara, M. Tachikawa, T. Tsukamoto, and T. Shimizu, “Observation of synchronization in laser chaos,” *Phys. Rev. Lett.* **72**, 3502 (1994).

⁶⁸A. Sawadsky, H. Kaufer, R. Moghadas Nia, S. P. Tarabrin, F. Y. Khalili, K. Hammerer, and R. Schnabel, “Observation of generalized optomechanical coupling and cooling on cavity resonance,” *Phys. Rev. Lett.* **114**, 043601 (2015).

⁶⁹T. M. Karg, B. Gouraud, C. Tat Ngai, G.-L. Schmid, K. Hammerer, and P. Treutlein, “Light-mediated strong coupling between a mechanical oscillator and atomic spins 1 meter apart,” *Science* **369**, 174 (2020).

⁷⁰J. von Neumann and E. Wigner, “Über merkwürdige diskrete Eigenwerte,” in *The Collected Works of Eugene Paul Wigner*, Volume A/1, edited by A. S. Wightman (Springer, Berlin, Heidelberg, 1993), pp. 291–293.

⁷¹C. W. Hsu, B. Zhen, A. Douglas Stone, J. D. Joannopoulos, and M. Soljčić, “Bound states in the continuum,” *Nat. Rev. Mater.* **1**, 16048 (2016).

⁷²J. N. Moreno, C. W. Wächter, and A. Eisfeld, “Synchronized states in dissipatively coupled harmonic oscillator networks,” *arXiv.2301.13614* (2023).